

TOAN VO

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EDUCATION

University of California, Irvine

October 2022 - June 2024

B.S. in Computer Science

GPA: 3.7/4.0

SKILLS

Programing Languages: C, C++, Java, Python, MySQL, PostgreSQL, JavaScript, React, Next.js

Technology: AWS, Tomcat, Firebase, VIM, Unix, Git, VSCode, IntelliJ, Arduino/STM32/Tiva/Raspberry Pi Microcontroller, Windows/Mac/Linux OS, MS Office, OpenAI, LTE/5G

Relevant Courses: Data Structures, Principles in System Design, Operating System, Data Management, Artificial Intelligence, Computer Network, Web Development, Embedded Software, Internet of Things, Compilers and Interpreter, Software Engineer

Languages: English, Vietnamese

PROJECTS

Full-Stack Coffee Shop Web Application

January 2025

- Developed a full-stack web application using Express.js, React.js, and PostgreSQL with full CRUD functionality.
- Integrated Firebase authentication for secure login and real-time database capabilities.
- Migrated the landing page to Next.js to improve SEO and performance using SSR and SSG.
- Deployed the project online, demonstrating scalability, debugging, and full-stack development skills.

Compiler Development

April 2024 – June 2024

- Built a compiler in Java 11 to translate a small imperative programming language into x86 assembly.
- Implemented five development stages: parsing Crux source code (ANTLR parse tree), generating an Abstract Syntax Tree (AST), type checking, intermediate representation, and assembly code generation.

AI-Powered Sudoku Solver

March 2024

- Implemented AI algorithms to solve expert-level 25x25 Sudoku boards in ~9 seconds.
- Utilized Forward Checking, Norvig's Heuristic, Minimum Remaining Value (MRV) Heuristic with Degree Heuristic (tie-breaker), and Least Constraining Value Heuristic algorithms for efficient solving.

Full-Stack Movie Website & Android Application

September 2023 – December 2023

- Wrote 7,706 lines of Java, JavaScript, HTML, CSS code to build a movie website and Android app, designing the architecture and setting up AWS, MySQL, and Tomcat. Developed an ETL pipeline for parsing large XML files, enriching a large movie database. Built a feature-rich frontend with JavaScript, jQuery, and AJAX, integrating with the Android app via a shared backend API.
- Implemented ~20 key features, including a movie catalog, secure login (SHA-256), full-text search with caching, cart checkout, bot detection (reCAPTCHA), and SQL injection protection. Optimized performance by 98% using MySQL connection pooling, replication, and Apache load balancing.

Postfix Calculator & NIM Game

September 2021 – December 2021

- Developed a postfix calculator in C++ capable of performing algebraic operations on big integers with unlimited size using the Chain of Responsibility pattern and stack data structure.
- Built a NIM game where players take turns removing stones from piles, applying structs and vector data structures for game logic.

EXPERIENCES

Network Engineer | BTI Solutions, Inc.

September 2024 – Present

- Verify network and software stability on the latest Android mobile products, including TCP/UDP communication testing.
- Communicate findings via technical reports and perform hardware/software maintenance, installations, and upgrades.
- Provide technical support for system development, troubleshooting, and field testing on 3GPP standards (GSM, EDGE, WCDMA, HSPA, LTE, 5G). Research, test, debug and monitor software patches, designing and executing test plans to analyze and resolve issues. Conduct TCP/UDP protocol analysis for network performance evaluation and optimization.
- Collaborate with R&D and QA teams to ensure issue resolution and support carrier lab testing as needed.